1. Identity

Document	STAR-PKG-STAR-TELEMETRY-EN · v2.2 · 2026-11-02
Package	star-telemetry
Version	0.2.7
Capability	star.telemetry.packets
Maturity	Released & Verified (RELEASED)
Python	⩾3.9
Source commit	dc09f61d
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_telemetry

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

NEEDS	LICENCE	SOURCE
star-timescales	Apache-2.0	S.T.A.R.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_telemetry-
```

```
0.2.7-py3-none-any.whl
```

FILE	SHA-256
star_telemetry-0.2.7-py3-none-any.whl	d4c3808f025685bd39a750a85ce8125022cdd268c10dc4a1ff403f3ac099400a
star_telemetry-0.2.7-source.zip	c49bec6b85abd27319e8a3daceb895502d436bdca2eb4f5cd5de6bd39584beeb

5. What it does, and its limits

CCSDS AOS v2 / TM v1 transfer-frame deframer and Space Packet reassembler, with CUC time decoding via star-timescales.

TC frames, USLP, frame synchronisation/ASM search, Reed-Solomon, the F Prime flight AosFramer component as oracle (test harness only), any flight qualification.

- `CcsdsTransferFrameEngine(frame_length, has_fecf).parse_frame(bytes) → (header, data_field)`: TFVN, SCID, VCID, VC frame count, replay flag, first header pointer, FECF (CRC-16, poly 0x1021, init 0xFFFF) validity.
- `process_raw_stream(bytes)` reassembles Space Packets across frames per virtual channel, including spanning packets. A packet is delivered in the frame that completes it. After a lost frame, the rest of the interrupted packet is discarded up to the next first header pointer. A pending packet whose size disagrees with its declared length is dropped, never cut to size.
- A frame of the wrong length is rejected with `ValueError`; a corrupted frame is reported (`fecf_valid = False`), never silently accepted.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_telemetry_0_2_7,
  author  = {Cavazzini, Andrea},
  title   = {{star-telemetry}: CCSDS AOS/TM transfer frame deframer and Space Packet
reassembler},
  version = {0.2.7},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_telemetry/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61d}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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